

The Hall–Jackson–Sudbery–Wild window is tight: a modular hyperbola in the $2p \times 2p$ grid $G(p)$ contains no more than $3(p - 1)$ points in general position

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with three autonomous Claude agents (Anthropic);
this note was written by the third agent — see the statement at the end

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Abstract

Let p be an odd prime, $c \in \mathbb{F}_p^*$, and let $H(c, p) = \{(x, y) \in \mathbb{Z}^2 : xy \equiv c \pmod{p}\}$ be the modular hyperbola. Hall, Jackson, Sudbery and Wild (1975) observed that the window $G(p) = [-\frac{p-1}{2}, \frac{3p-1}{2}] \times [0, 2p-1]$ contains exactly four points of each of the $p-1$ residue classes of $H(c, p)$, at the vertices of a $p \times p$ square, and that keeping three of the four (the twelve outer blocks T_2 of the 4×4 block subdivision of $G(p)$) yields $3(p-1)$ points with no three collinear. We prove that this is optimal: *every* subset of $H(c, p) \cap G(p)$ with no three collinear points has at most $3(p-1)$ points. The proof is a double count over the lines of slope ± 1 that carry three or four points of $H(c, p) \cap G(p)$. The argument also determines all maximum sets: their number is 9^s , where $s \in \{0, 1, 2\}$ is the number of quadratic residues among c and $-c$; for $s = 0$ the HJSW set $H(c, p) \cap T_2$ is the unique maximum. Everything is verified by an independent program for all primes $p \leq 101$.

1 Statement

Throughout, p is an odd prime, $h = (p-1)/2$, $c \in \mathbb{F}_p^*$, and

$$H(c, p) = \{(x, y) \in \mathbb{Z}^2 : xy \equiv c \pmod{p}\}, \quad G(p) = \{-h, \dots, 3h+1\} \times \{0, \dots, 2p-1\},$$

$$\mathcal{P} := H(c, p) \cap G(p).$$

$G(p)$ is a $2p \times 2p$ grid (this is the grid G of [2, (2.1)], taken from [1]). A set is *lawful* (in general position, no-three-in-line) if it contains no three collinear points.

Two points are *congruent* if their coordinates agree modulo p ; a *class* is a congruence class, identified with an element of $\mathbb{F}_p \times \mathbb{F}_p$. Points of $H(c, p)$ have both coordinates prime to p , so the classes of $H(c, p)$ are the $p-1$ pairs $\kappa = (a, b) \in \mathbb{F}_p^* \times \mathbb{F}_p^*$ with $ab = c$. Since each coordinate range of $G(p)$ is an interval of $2p$ consecutive integers, it contains exactly two representatives of every residue class; hence every class κ of $H(c, p)$ has exactly four points in $G(p)$ and $|\mathcal{P}| = 4(p-1)$. We write (x_κ, y_κ) for the *base copy* of κ , its unique point in $\{-h, \dots, h\} \times \{1, \dots, p-1\}$ (necessarily $x_\kappa \neq 0$), and

$$\kappa(r, s) := (x_\kappa + rp, y_\kappa + sp), \quad r, s \in \{0, 1\},$$

for its four points in $G(p)$. Following [1, 2] we sort the classes into four *types* according to the position of the base copy:

$$A : x_\kappa > 0, y_\kappa > h; \quad B : x_\kappa < 0, y_\kappa > h; \quad C : x_\kappa > 0, y_\kappa \leq h; \quad D : x_\kappa < 0, y_\kappa \leq h.$$

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(In the notation of [2, Def. 2.3]: $\kappa(0,0) \in A_{0,0}, B_{-1,0}, C_{0,0}, D_{-1,0}$ respectively; the HJSW set T_2 omits, for each type, the copy $A_{0,0}, B_{0,0}, C_{0,1}, D_{0,1}$, i.e. $\kappa(0,0)$ for A , $\kappa(1,0)$ for B , $\kappa(0,1)$ for C , $\kappa(1,1)$ for D ; the union of these omitted blocks is the middle block M of [2, §2.3].) We write n_A, n_B, n_C, n_D for the numbers of classes of each type, so $n_A + n_B + n_C + n_D = p - 1$. Finally, for a class κ put

$$d_\kappa := x_\kappa - y_\kappa, \quad e_\kappa := x_\kappa + y_\kappa.$$

Three involutions act on the classes of $H(c, p)$:

$$\sigma(a, b) = (-b, -a), \quad \tau(a, b) = (b, a), \quad \nu(a, b) = (-a, -b) = \sigma\tau(a, b) = \tau\sigma(a, b).$$

σ has fixed classes iff $-c$ is a quadratic residue (then exactly two: $(a, -a)$ with $a^2 = -c$), τ has fixed classes iff c is a quadratic residue (two: (a, a) with $a^2 = c$), and ν has none. Put

$$f_\sigma := \#\{\kappa : \sigma\kappa = \kappa\} \in \{0, 2\}, \quad f_\tau := \#\{\kappa : \tau\kappa = \kappa\} \in \{0, 2\},$$

$$s := \frac{1}{2}(f_\sigma + f_\tau) = [-c \text{ is a QR}] + [c \text{ is a QR}] \in \{0, 1, 2\}.$$

Theorem 1. *Every lawful subset of $\mathcal{P} = H(c, p) \cap G(p)$ has at most $3(p - 1)$ points. The bound is attained, and the number of lawful subsets of size $3(p - 1)$ is exactly 9^s . In particular, if neither c nor $-c$ is a quadratic residue modulo p , the HJSW set $H(c, p) \cap T_2$ is the unique maximum lawful subset of $\mathcal{P}$.*

For $c = 1$: there are 81 maximum sets if $p \equiv 1 \pmod{4}$ and 9 if $p \equiv 3 \pmod{4}$; all of them agree with the HJSW set outside the two or four classes $(\pm a, \mp a)$, $(\pm a, \pm a)$ with $a^2 = \mp 1$.

2 Lines through three points of $\mathcal{P}$

Lemma 2. *Let ℓ be a line containing at least two points of $H(c, p)$. Then the points of $H(c, p) \cap \ell$ lie in at most two classes. More precisely, if ℓ contains a point of class κ , then the classes met by ℓ are: only κ if ℓ has slope 0 or ∞ ; among $\{\kappa, \sigma\kappa\}$ if ℓ has slope $+1$; among $\{\kappa, \tau\kappa\}$ if ℓ has slope -1 .*

Proof. Let $(u, v) \in \mathbb{Z}^2$ be a primitive direction vector of ℓ and $(x_0, y_0) \in H(c, p) \cap \ell$. The lattice points of ℓ are $(x_0 + tu, y_0 + tv)$, $t \in \mathbb{Z}$, and such a point lies on $H(c, p)$ iff $uv t^2 + (x_0 v + y_0 u) t \equiv 0 \pmod{p}$ (we used $x_0 y_0 \equiv c$). The class of the point depends only on $t \pmod{p}$. If $uv \not\equiv 0$, the congruence has at most two roots modulo p ; if exactly one of u, v is divisible by p (both cannot be, as $\gcd(u, v) = 1$), it reduces to $x_0 v t \equiv 0$ or $y_0 u t \equiv 0$ with a unit coefficient, so $t \equiv 0$: one class. For $(u, v) = (1, 0), (0, 1)$ this gives the claim for slopes 0, ∞ . For $(u, v) = (1, 1)$ the roots are $t \equiv 0$ and $t \equiv -(x_0 + y_0)$, the latter giving the point $(-y_0, -x_0)$ modulo p , of class $\sigma\kappa$. For $(u, v) = (1, -1)$ the roots are $t \equiv 0$ and $t \equiv y_0 - x_0$, giving (y_0, x_0) , of class $\tau\kappa$. $\square$

Lemma 3. *Two distinct points of one class in $G(p)$ differ by $\pm(p, 0)$, $\pm(0, p)$, $\pm(p, p)$ or $\pm(p, -p)$; the line through them has slope 0, ∞ , $+1$ or -1 and contains no third point of that class. The point $\kappa(r, s)$ lies on the slope- $(+1)$ line $x - y = d_\kappa + (r - s)p$ and on the slope- (-1) line $x + y = e_\kappa + (r + s)p$.*

Proof. The four points of a class are the vertices of an axis-parallel square of side p ; no three are collinear. The last sentence is a direct computation. $\square$

Corollary 4. *A line containing at least three points of $\mathcal{P}$ has slope $+1$ or -1 .*

Proof. By Lemma 2 the three points lie in at most two classes, so two of them are congruent, and by Lemma 3 the line has slope 0, ∞ , ± 1 . A line of slope 0 or ∞ meets a single class (Lemma 2), which has at most two points on it (Lemma 3). $\square$

Lemma 5 (position of the partner classes). *Let κ be a class of $H(c, p)$.*

(i) *If $\kappa \in A$ then $\sigma\kappa \in A$ and $d_{\sigma\kappa} = d_\kappa$; if $\kappa \in D$ then $\sigma\kappa \in D$ and $d_{\sigma\kappa} = d_\kappa$; if $\kappa \in C$ then $\sigma\kappa \in B$ and $d_{\sigma\kappa} = d_\kappa - p$; if $\kappa \in B$ then $\sigma\kappa \in C$ and $d_{\sigma\kappa} = d_\kappa + p$.*

(ii) *If $\kappa \in B$ then $\tau\kappa \in B$ and $e_{\tau\kappa} = e_\kappa$; if $\kappa \in C$ then $\tau\kappa \in C$ and $e_{\tau\kappa} = e_\kappa$; if $\kappa \in A$ then $\tau\kappa \in D$ and $e_{\tau\kappa} = e_\kappa - p$; if $\kappa \in D$ then $\tau\kappa \in A$ and $e_{\tau\kappa} = e_\kappa + p$.*

In particular σ preserves the types A, D and interchanges B, C ; τ preserves B, C and interchanges A, D ; the fixed classes of σ lie in $A \cup D$ and those of τ in $B \cup C$; and $n_A = n_D, n_B = n_C$.

Proof. Write $\kappa = (a, b)$ with base copy (x_κ, y_κ) , so $x_\kappa \equiv a, y_\kappa \equiv b$. The base copy of $\sigma\kappa = (-b, -a)$ has first coordinate $-y_\kappa$ if $y_\kappa \leq h$ and $p - y_\kappa$ if $y_\kappa > h$, and second coordinate $p - x_\kappa$ if $x_\kappa > 0$ and $-x_\kappa$ if $x_\kappa < 0$ (these are the representatives of $-b$ in $[-h, h]$ and of $-a$ in $[1, p-1]$). Going through the four types: $\kappa \in A$: $(p - y_\kappa, p - x_\kappa)$, again of type A , difference $x_\kappa - y_\kappa$; $\kappa \in D$: $(-y_\kappa, -x_\kappa)$, type D , difference $x_\kappa - y_\kappa$; $\kappa \in C$: $(-y_\kappa, p - x_\kappa)$, type B , difference $x_\kappa - y_\kappa - p$; $\kappa \in B$: $(p - y_\kappa, -x_\kappa)$, type C , difference $x_\kappa - y_\kappa + p$. Similarly, the base copy of $\tau\kappa = (b, a)$ has first coordinate y_κ if $y_\kappa \leq h$ and $y_\kappa - p$ otherwise, and second coordinate x_κ if $x_\kappa > 0$ and $x_\kappa + p$ otherwise: $\kappa \in B$: $(y_\kappa - p, x_\kappa + p)$, type B , sum e_κ ; $\kappa \in C$: (y_κ, x_κ) , type C , sum e_κ ; $\kappa \in A$: $(y_\kappa - p, x_\kappa)$, type D , sum $e_\kappa - p$; $\kappa \in D$: $(y_\kappa, x_\kappa + p)$, type A , sum $e_\kappa + p$. The remaining assertions follow ($\nu = \sigma\tau$ is a fixed-point-free involution interchanging $A \leftrightarrow D$ and $B \leftrightarrow C$). $\square$

Let $\mathcal{L}$ be the set of lines containing at least three points of $\mathcal{P}$. For a class κ let

$$\begin{aligned} D_\kappa &:= \{x - y = d_\kappa\} \quad (\text{the diagonal of } \kappa; \text{ it contains } \kappa(0, 0), \kappa(1, 1)), \\ E_\kappa &:= \{x + y = e_\kappa + p\} \quad (\text{the antidiagonal of } \kappa; \text{ it contains } \kappa(0, 1), \kappa(1, 0)). \end{aligned}$$

Proposition 6. $\mathcal{L}$ consists of the following lines, and no others; for each we list its points in $\mathcal{P}$.

- (a) For $\kappa \in A \cup D$ with $\sigma\kappa \neq \kappa$: the common diagonal $D_\kappa = D_{\sigma\kappa}$, with the four points $\kappa(0, 0), \kappa(1, 1), \sigma\kappa(0, 0), \sigma\kappa(1, 1)$.
- (b) For $\kappa \in B \cup C$ with $\tau\kappa \neq \kappa$: the common antidiagonal $E_\kappa = E_{\tau\kappa}$, with the four points $\kappa(0, 1), \kappa(1, 0), \tau\kappa(0, 1), \tau\kappa(1, 0)$.
- (c) For every $\kappa \in C$: D_κ with the three points $\kappa(0, 0), \kappa(1, 1), \sigma\kappa(1, 0)$ ($\sigma\kappa \in B$); for every $\kappa \in B$: D_κ with the three points $\kappa(0, 0), \kappa(1, 1), \sigma\kappa(0, 1)$ ($\sigma\kappa \in C$).
- (d) For every $\kappa \in A$: E_κ with the three points $\kappa(0, 1), \kappa(1, 0), \tau\kappa(1, 1)$ ($\tau\kappa \in D$); for every $\kappa \in D$: E_κ with the three points $\kappa(0, 1), \kappa(1, 0), \tau\kappa(0, 0)$ ($\tau\kappa \in A$).

Consequently

$$|\mathcal{L}| = \frac{n_A + n_D - f_\sigma}{2} + \frac{n_B + n_C - f_\tau}{2} + (n_A + n_D) + (n_B + n_C) = \frac{3}{2}(p-1) - \frac{f_\sigma + f_\tau}{2}. \quad (1)$$

Moreover, the points of $\mathcal{P}$ lying on no line of $\mathcal{L}$ are exactly $\kappa(1, 1)$ for σ -fixed $\kappa \in A$, $\kappa(0, 0)$ for σ -fixed $\kappa \in D$, $\kappa(0, 1)$ for τ -fixed $\kappa \in B$ and $\kappa(1, 0)$ for τ -fixed $\kappa \in C$; their number is

$$Z := \#\{q \in \mathcal{P} : q \text{ lies on no line of } \mathcal{L}\} = f_\sigma + f_\tau. \quad (2)$$

The points lying on two lines of $\mathcal{L}$ are exactly the “middle-block” copies $\kappa(0, 0)$ ($\kappa \in A$), $\kappa(1, 0)$ ($\kappa \in B$), $\kappa(0, 1)$ ($\kappa \in C$), $\kappa(1, 1)$ ($\kappa \in D$) of the classes κ whose partner ($\sigma\kappa$ for A, D ; $\tau\kappa$ for B, C) is different from κ ; no point lies on three lines of $\mathcal{L}$.

Proof. Let $\ell \in \mathcal{L}$. By Corollary 4 its slope is ± 1 ; say $+1$ (the case -1 is symmetric, with τ, e_κ, E_κ in place of $\sigma, d_\kappa, D_\kappa$). By Lemma 2 the points of $\ell \cap \mathcal{P}$ lie in two classes $\kappa, \sigma\kappa$ (a single class carries at most two points on a line), and one of the classes, say κ , has two points on ℓ ; by Lemma 3 these are $\kappa(0,0), \kappa(1,1)$ and $\ell = D_\kappa$, and $\sigma\kappa \neq \kappa$. The points of $\sigma\kappa$ on D_κ are the $\sigma\kappa(r,s)$ with $d_{\sigma\kappa} + (r-s)p = d_\kappa$. By Lemma 5(i): for $\kappa \in A \cup D$, $r = s$, giving (a); for $\kappa \in C$, $r - s = 1$, giving $\sigma\kappa(1,0)$; for $\kappa \in B$, $r - s = -1$, giving $\sigma\kappa(0,1)$; this is (c). Conversely each listed line has ≥ 3 points. Note that for $\kappa \in B \cup C$ the diagonals D_κ and $D_{\sigma\kappa}$ are distinct lines, each in $\mathcal{L}$, whereas for $\kappa \in A \cup D$ they coincide. Counting: (a) gives $(n_A + n_D - f_\sigma)/2$ lines, (b) $(n_B + n_C - f_\tau)/2$, (c) $n_B + n_C$, (d) $n_A + n_D$, which is (1).

For the incidences of a single point, take $\kappa \in A$. Its points $\kappa(0,1), \kappa(1,0)$ lie on $E_\kappa \in \mathcal{L}$ (d), and on no line of type (a)/(c) (their slope- $(+1)$ lines $x - y = d_\kappa \pm p$ carry no second point of κ and, by (i), no point of $\sigma\kappa$). $\kappa(1,1)$ lies on D_κ , which is in $\mathcal{L}$ iff $\sigma\kappa \neq \kappa$, and on $x + y = e_\kappa + 2p = e_{\tau\kappa} + 3p$, which carries no point of $\tau\kappa$: so $\kappa(1,1)$ is on one line of $\mathcal{L}$ if $\sigma\kappa \neq \kappa$ and on none otherwise. $\kappa(0,0)$ lies on D_κ and on $x + y = e_\kappa = e_{\tau\kappa} + p$, i.e. on $E_{\tau\kappa} \in \mathcal{L}$: two lines if $\sigma\kappa \neq \kappa$, one otherwise. The types D, B, C are analogous (for B : $\kappa(1,0) \in D_{\sigma\kappa}$ since $d_\kappa + p = d_{\sigma\kappa}$, while $\kappa(0,1)$ lies on $x - y = d_\kappa - p = d_{\sigma\kappa} - 2p$, which carries no point of $\sigma\kappa$; for C the roles of $(1,0), (0,1)$ are exchanged; for D : $\kappa(1,1) \in E_{\tau\kappa}$ since $e_\kappa + 2p = e_{\tau\kappa} + p$, and $\kappa(0,0)$ lies on $x + y = e_\kappa = e_{\tau\kappa} - p$). This gives (2) and the last two assertions. $\square$

3 Proof of Theorem 1

Let $S \subseteq \mathcal{P}$ be lawful. For $q \in \mathcal{P}$ let $m(q)$ be the number of lines of $\mathcal{L}$ through q . Every line of $\mathcal{L}$ carries at most two points of S , hence

$$\begin{aligned} |S| &= \#\{q \in S : m(q) = 0\} + \#\{q \in S : m(q) \geq 1\} \leq Z + \sum_{q \in S} m(q) - \sum_{q \in S} (m(q) - 1)^+ \\ &= Z + \sum_{\ell \in \mathcal{L}} |S \cap \ell| - \sum_{q \in S} (m(q) - 1)^+ \leq Z + 2|\mathcal{L}|, \end{aligned} \tag{3}$$

and by (1), (2)

$$Z + 2|\mathcal{L}| = (f_\sigma + f_\tau) + 3(p-1) - (f_\sigma + f_\tau) = 3(p-1).$$

This proves the bound. (Lawfulness of S was used only through the lines of $\mathcal{L}$; conversely, since every line carrying three points of $\mathcal{P}$ belongs to $\mathcal{L}$, a set $S \subseteq \mathcal{P}$ is lawful iff $|S \cap \ell| \leq 2$ for all $\ell \in \mathcal{L}$.)

Equality. $|S| = 3(p-1)$ holds iff all inequalities in (3) are equalities, i.e. iff (1) S contains all Z points lying on no line of $\mathcal{L}$; (2) $|S \cap \ell| = 2$ for every $\ell \in \mathcal{L}$; (3) no point of S lies on two lines of $\mathcal{L}$. Group the classes into orbits of the Klein four-group $\{1, \sigma, \tau, \nu\}$. A generic orbit $\{\kappa, \sigma\kappa, \tau\kappa, \nu\kappa\}$ has four distinct classes, all with distinct partners; the six lines of $\mathcal{L}$ meeting it (Proposition 6) meet no other orbit, its four middle-block copies have $m = 2$ and are excluded by (3), and each of the six lines then retains exactly two points with $m = 1$, which are forced into S by (2): on this orbit S coincides with $H(c, p) \cap T_2$. A non-generic orbit is $\{\kappa, \nu\kappa\}$ with κ (and $\nu\kappa$) fixed by σ or by τ ; it meets exactly two lines of $\mathcal{L}$, each with three points of $m = 1$, plus two points with $m = 0$: conditions (1)–(3) allow $3 \cdot 3 = 9$ choices, all lawful. There are $s = (f_\sigma + f_\tau)/2$ non-generic orbits, whence 9^s maximum sets. $\square$

Remark 7. (a) The count (3) is a linear-programming certificate: the dual solution assigns weight 1 to every line of $\mathcal{L}$ (constraint $\sum_{q \in \ell} x_q \leq 2$) and weight 1 to the upper bound $x_q \leq 1$ of each of the Z unconstrained points; so even the fractional relaxation of the problem has value $3(p-1)$.

(b) The theorem says that no lawful set can use the middle block M profitably: adding fourth copies to the HJSW set is impossible without removing at least as many points. It complements Observations 2.4–2.6 of [2] (which yield the lower bound $3(p-1)$) and settles, for a

single hyperbola in its own window, the question raised there whether the HJSW construction can be improved: within $H(c, p) \cap G(p)$ it cannot; any improvement must use points outside $H(c, p) \cap G(p)$ (another hyperbola, another window, or non-hyperbola points).

(c) The proof uses only: at most two classes per line (Lagrange), the square arrangement of the four copies, and the position of the partner classes $\sigma\kappa$, $\tau\kappa$ (Lemma 5). It does not use $p \geq 7$; $p = 3$ and $p = 5$ are covered as well.

4 Computer verification

The script `slack/hjsw_window_check.py` of the repository recomputes, from the point set alone (no part of the proof is trusted): all lines with at least three points of $\mathcal{P}$ (all have slope ± 1 , three or four points, and lie inside one orbit of classes); the identity $2|\mathcal{L}| + Z = 3(p - 1)$; the exact maximum by exhaustive search inside each orbit (at most 2^{16} subsets), assembling a maximum witness which is passed through the independent verifier `saturation.the_law` in grid coordinates; and the number of maximum sets, compared with 9^s . All checks pass for every prime $3 \leq p \leq 101$ (all $c \in \mathbb{F}_p^*$ for $p \leq 31$; $c \in \{1, 2, 3, -1, -2, (p+1)/2\}$ for larger p); log: `bench/hjsw_window_check_all.log`. The values agree with the exact maxima obtained earlier by SAT (kissat/CaDiCaL, two encodings, witnesses through the law): 18, 30, 36, 48, 54, 66 for $p = 7, 11, 13, 17, 19, 23$.

p	c	$ \mathcal{P} $	$ \mathcal{L} $	Z	max	#max sets
7	1	24	8	2	18	9
11	1	40	14	2	30	9
13	1	48	16	4	36	81
13	2	48	18	0	36	1
17	1	64	22	4	48	81
19	1	72	26	2	54	9
23	1	88	32	2	66	9
101	1	400	148	4	300	81

Table 1: Sample values from the verification log ($|\mathcal{L}| = \frac{3}{2}(p - 1) - s$, $Z = 2s$).

Statement on authorship. The theorem, its proof and the verification program were produced by an autonomous Claude agent (the third agent of the project) working from the setup and the computational observations of the first two agents (exact maxima by SAT for $p \leq 23$); the owner set the question. Everything is reproducible from the repository.

References

- [1] R. R. Hall, T. H. Jackson, A. Sudbery, K. Wild, Some advances in the no-three-in-line problem, *J. Combin. Theory Ser. A* 18 (1975) 336–341.
- [2] B. Kovács, Z. L. Nagy, D. R. Szabó, Randomised algebraic constructions for the no- $(k + 1)$ -in-line problem, *arXiv:2508.07632* (2025); *Advances in Combinatorics* 2026:7.